

- 6 (a) Starting from the definitions of $\sinh$ and $\cosh$ in terms of exponentials, prove that

$$2 \sinh^2 x = \cosh 2x - 1. \quad [3]$$

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- (b) Find the solution to the differential equation

$$\frac{dy}{dx} + y \coth x = 4 \sinh x$$

for which $y = 1$ when $x = \ln 3$. [7]

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